

# EE3621: Digital Signal Processing

## Student Lecture Notes — Lecture 21

### Linear Phase FIR Filters, Characteristics, Zeros & Moving Average Filters

#### III B.Tech EEE — Core Exam Preparation Guide

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## 1 Core Definitions & Linear Phase Conditions

### 1.1 Distortionless Transmission & Phase Delay vs. Group Delay

An LTI filter transmits an input signal without waveform distortion if all frequency components experience:

1. A constant gain:  $|H(e^{j\omega})| = A_0$
  2. A strictly constant time delay:  $\tau_g = \alpha$
- **Phase Delay:** The time delay experienced by each individual sinusoidal carrier:

$$\tau_p(\omega) = -\frac{\theta(\omega)}{\omega} \quad (1)$$

- **Group Delay:** The time delay experienced by the signal's modulation envelope:

$$\tau_g(\omega) = -\frac{d\theta(\omega)}{d\omega} \quad (2)$$

**The Linear Phase Criterion:** If  $\theta(\omega) = -\alpha\omega$ , then  $\tau_p(\omega) = \tau_g(\omega) = \alpha = \text{constant}$ . This eliminates **phase distortion** and **envelope dispersion**.

### 1.2 Time-Domain Conditions for Linear Phase

For a causal FIR filter of length  $N$  (coefficients  $h[0], h[1], \dots, h[N-1]$ ):

- **Strict Linear Phase** ( $\theta(\omega) = -\alpha\omega$ ): Requires **even symmetry**:

$$h[n] = h[N-1-n], \quad \alpha = \frac{N-1}{2} \quad (3)$$

- **Generalized Linear Phase** ( $\theta(\omega) = \beta - \alpha\omega$ , where  $\beta = \pm\pi/2$ ): Requires **odd anti-symmetry**:

$$h[n] = -h[N-1-n], \quad \alpha = \frac{N-1}{2} \quad (4)$$

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## 2 The 4 Linear-Phase FIR Filter Types (Master Reference)

Every linear-phase FIR filter belongs to one of four distinct classes based on coefficient symmetry and length parity:

Table 1: Classification and Filter Suitability of Linear-Phase FIR Filters

| Type   | Symmetry                             | Length $N$      | Phase $\theta(\omega)$          | Mandatory Zeros                   | Filter Suitability             |
|--------|--------------------------------------|-----------------|---------------------------------|-----------------------------------|--------------------------------|
| Type 1 | Symmetric ( $h[n] = h[N-1-n]$ )      | Odd ( $M$ even) | $-\alpha\omega$                 | None                              | Universal: LPF, HPF, BPF, BSF  |
| Type 2 | Symmetric ( $h[n] = h[N-1-n]$ )      | Even ( $M$ odd) | $-\alpha\omega$                 | $\omega = \pi$ ( $z = -1$ )       | LPF, BPF only (Never HPF/BSF!) |
| Type 3 | Antisymmetric ( $h[n] = -h[N-1-n]$ ) | Odd ( $M$ even) | $\frac{\pi}{2} - \alpha\omega$  | $\omega = 0, \pi$ ( $z = \pm 1$ ) | BPF only (Never LPF/HPF/BSF)   |
| Type 4 | Antisymmetric ( $h[n] = -h[N-1-n]$ ) | Even ( $M$ odd) | $\frac{3\pi}{2} - \alpha\omega$ | $\omega = 0$ ( $z = 1$ )          | HPF, BPF, Differentiator       |

### Critical Exam Insights:

- **Why Type 2 cannot be Highpass:** At  $\omega = \pi$ ,  $e^{j\pi(n-\alpha)} = \cos(\pi(n-\alpha))$ . For  $N$  even,  $\alpha = (N-1)/2$  is a half-integer, making  $\cos(\pi(n-\alpha)) = 0$ . Hence,  $\mathbf{H}(\mathbf{e}^{j\pi}) \equiv \mathbf{0}$  unconditionally!
- **Why Antisymmetric Filters (Types 3 & 4) have zero at  $\omega = 0$ :** Since  $h[n] = -h[N-1-n]$ , setting  $\omega = 0$  gives  $H(e^{j0}) = \sum h[n] = 0$ . They block DC completely.

## 3 Location of Zeros in the Complex Z-Plane

Linear phase imposes reciprocal symmetry on the system function:

$$H(z) = \pm z^{-(N-1)} H(z^{-1}) \quad (5)$$

Consequently, if  $z_0 = re^{j\phi}$  is a zero of  $H(z)$ , its reciprocal  $1/z_0 = \frac{1}{r}e^{-j\phi}$  must also be a zero. For real impulse responses, complex zeros must also occur in conjugate pairs.

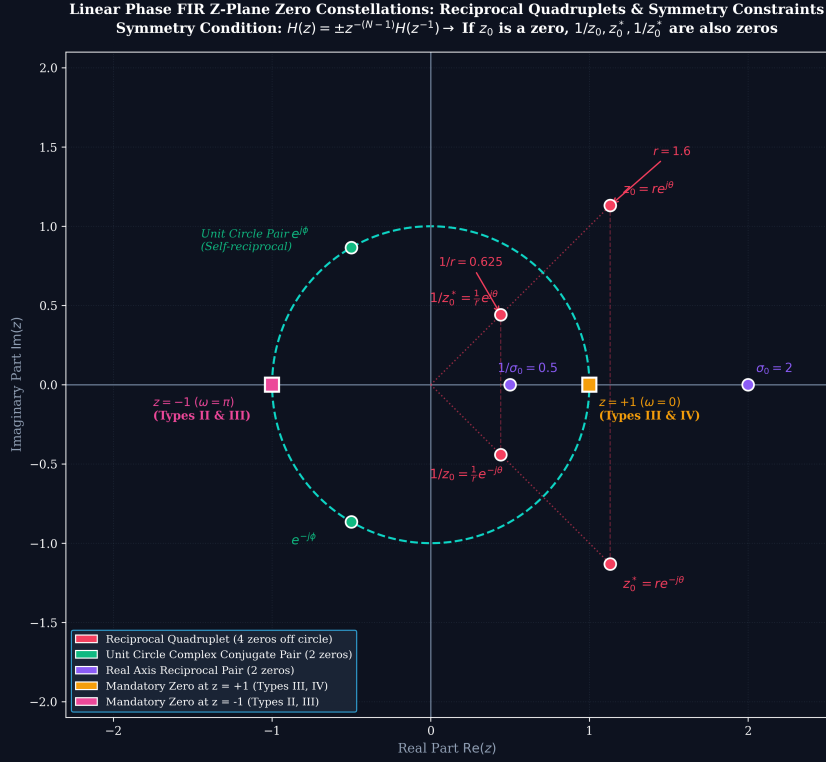


Figure 1: Zero constellations for linear-phase FIR systems: complex quadruplets off the unit circle, pairs on the unit circle ( $|z| = 1$ ), and mandatory boundary zeros at  $z = \pm 1$ .

- **Complex zero off the unit circle ( $|z| \neq 1$ ):** Appears as a **reciprocal conjugate quadruplet of 4 zeros**:

$$\left\{ re^{j\phi}, \quad re^{-j\phi}, \quad \frac{1}{r}e^{j\phi}, \quad \frac{1}{r}e^{-j\phi} \right\} \quad (6)$$

- **Complex zero on the unit circle ( $|z| = 1$ ):** Since  $1/e^{j\phi} = e^{-j\phi}$ , the zero is its own reciprocal. It occurs as a **conjugate pair of 2 zeros**:

$$\left\{ e^{j\phi}, \quad e^{-j\phi} \right\} \quad (7)$$

These zeros create sharp transmission nulls in the filter's stopband.

- **Real zero off the unit circle:** Occurs as a **reciprocal pair of 2 zeros**:  $\{\sigma, 1/\sigma\}$ .
- **Real zero on the unit circle:** A single zero at  $z = 1$  ( $\omega = 0$ ) or  $z = -1$  ( $\omega = \pi$ ).

## 4 Moving Average (MA) Filters

### 4.1 Basic Equations

An  $M$ -point Moving Average filter is defined by:

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n-k] \iff h[n] = \frac{1}{M}, \quad 0 \leq n \leq M-1 \quad (8)$$

- **Symmetry:**  $h[n] = h[M - 1 - n] = \frac{1}{M} \implies$  **Always Linear Phase** with group delay  $\tau = \frac{M-1}{2}$ .

- **Frequency Response:**

$$H(e^{j\omega}) = \left[ \frac{\sin(M\omega/2)}{M \sin(\omega/2)} \right] e^{-j(\frac{M-1}{2})\omega} \quad (9)$$

- **Zeros in Z-Domain:**  $H(z) = \frac{1}{M} \frac{1-z^{-M}}{1-z^{-1}}$ . The numerator has zeros at the  $M$  roots of unity  $e^{j2\pi k/M}$ . The zero at  $k = 0$  ( $z = 1$ ) is cancelled by the denominator pole at  $z = 1$ .
- Exactly  $M - 1$  zeros lie on the unit circle:  $\mathbf{z}_k = e^{j\frac{2\pi k}{M}}$  for  $k = 1, 2, \dots, M - 1$ , producing transmission notches at harmonic frequencies.

## 5 Standard Solved Exam Example

**Problem:** An FIR filter has the impulse response:

$$h[n] = \{1, 2, 3, 2, 1\}, \quad \text{for } n = 0, 1, 2, 3, 4 \quad (10)$$

1. Verify linear phase and determine the group delay  $\tau_g$ .
2. Identify the filter type (Type 1 to 4) and its filter suitability.
3. Find the amplitude function  $A(\omega)$  and evaluate the DC gain  $H(e^{j0})$ .
4. Find all zeros of  $H(z)$  and sketch their constellation.

**Solution:**

**Step 1: Symmetry and Group Delay** Length  $N = 5$  (Odd). Center of symmetry:  $\alpha = \frac{N-1}{2} = \frac{5-1}{2} = 2$ . Checking symmetry:  $h[0] = h[4] = 1$ ,  $h[1] = h[3] = 2$ ,  $h[2] = 3$ . Since  $h[n] = h[4 - n]$ , the filter has **even symmetry** with constant group delay:

$$\tau_g = \alpha = 2 \text{ samples} \quad (11)$$

**Step 2: Filter Classification** Even symmetry + Odd length  $N = 5 \implies$  **Type 1 Linear-Phase FIR Filter**. Suitability: Universal (Can implement Lowpass, Highpass, Bandpass, or Bandstop filters).

**Step 3: Amplitude Response and DC Gain** Factoring out the linear-phase factor  $e^{-j2\omega}$ :

$$\begin{aligned} H(e^{j\omega}) &= e^{-j2\omega} [h[2] + 2h[1] \cos(\omega) + 2h[0] \cos(2\omega)] \\ &= e^{-j2\omega} [3 + 4 \cos(\omega) + 2 \cos(2\omega)] \end{aligned} \quad (12)$$

Thus:

$$\mathbf{A}(\omega) = 3 + 4 \cos(\omega) + 2 \cos(2\omega) \quad (13)$$

At DC ( $\omega = 0$ ):  $A(0) = 3 + 4(1) + 2(1) = 9 = \sum_{n=0}^4 h[n]$ .

**Step 4: Algebraic Zero Extraction**

$$H(z) = 1 + 2z^{-1} + 3z^{-2} + 2z^{-3} + z^{-4} = 0 \implies z^4 + 2z^3 + 3z^2 + 2z + 1 = 0 \quad (14)$$

Dividing by  $z^2$ :  $(z^2 + z^{-2}) + 2(z + z^{-1}) + 3 = 0$ . Let  $u = z + z^{-1}$  (so  $z^2 + z^{-2} = u^2 - 2$ ):

$$(u^2 - 2) + 2u + 3 = 0 \implies u^2 + 2u + 1 = 0 \implies (u + 1)^2 = 0 \implies u = -1 \text{ (multiplicity 2)} \quad (15)$$

Now solve  $z + z^{-1} = -1 \implies z^2 + z + 1 = 0$ :

$$z = \frac{-1 \pm j\sqrt{3}}{2} = e^{\pm j\frac{2\pi}{3}} \quad (\text{each of multiplicity 2}) \quad (16)$$

**Conclusion:** All 4 zeros lie on the unit circle at angles  $\pm 120^\circ$  (multiplicity 2), satisfying the reciprocal property  $|z| = 1$ .

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## 6 Essential Exam Checklist & Pitfalls

- ☐ **Group Delay Formula:** Always use  $\tau = \frac{N-1}{2}$ , **NOT**  $N/2$ .
- ☐ **Highpass Filter Rule:** Never select an even-length symmetric filter (Type 2) for a Highpass filter because  $H(e^{j\pi}) \equiv 0$ .
- ☐ **DC Zero Rule:** Never select an antisymmetric filter (Type 3 or Type 4) for a Lowpass filter because  $H(e^{j0}) \equiv 0$ .
- ☐ **Reciprocal Zero Pairing:** Always ensure that if  $z_0$  is a zero,  $1/z_0$  and  $z_0^*$  are also zeros.